Lecture 24 2025-12-04

General (explicit) s-stage Runge-Kutta method

$$(\Delta)\begin{cases} k_\ell = f\left(t_i + c_\ell h,\ y_i + h\sum_{j=1}^{\ell-1} a_{\ell j} k_j\right) \\ y_{i+1} = y_i + h\sum_{\ell=1}^{s} b_\ell k_\ell \end{cases}$$

k_ℓ = ℓth stage derivative, $c_\ell, a_{\ell j}, b_\ell$ = scheme coefficients

Usually represented in a Butcher array/tableau:

c	A
	b^T

Examples: Forward Euler:

0	0
	1

Midpoint rule:

0	0	0
$\frac{1}{2}$	$\frac{1}{2}$	0
	0	1

Modified Euler:

0	0	
1	1	0
	$\frac{1}{2}$	$\frac{1}{2}$

Heun's Method

0	0		
$\frac{1}{3}$	$\frac{1}{3}$	0	
$\frac{2}{3}$	0	$\frac{2}{3}$	0
	$\frac{1}{4}$	0	$\frac{3}{4}$

HW: Compare to definition of Heun's method in textbook

"the" Runge-Kutta (4th order accurate)

→ 0	0			
→ $\frac{1}{2}$	$\frac{1}{2}$	0		← A
c → $\frac{1}{2}$	0	$\frac{1}{2}$	0	
1	0	0	1	0
	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{2}{6}$	$\frac{1}{6}$ ← b

$$\rightarrow\ k_1 = f(t_i, y_i)$$

$$\rightarrow\ k_2 = f\left(t_i + \tfrac{1}{2}h,\ y_i + \tfrac{1}{2}h\,k_1\right)$$

$$k_3 = f\left(t_i + \tfrac{1}{2}h,\ y_i + \tfrac{1}{2}h\,k_2\right)$$

$$k_4 = f(t_i + h,\ y_i + h\,k_3)$$

$$y_{i+1} = y_i + h\left(\tfrac{1}{6}k_1 + \tfrac{1}{3}k_2 + \tfrac{1}{3}k_3 + \tfrac{1}{6}k_4\right)$$

Truncation error $h\tau$ = "what is left when plugging in the exact solution in (Δ)"

"1 step error"

Let's analyze the general explicit 2-stage method

$$\begin{array}{c|cc} 0 & 0 & \\ a & a & 0 \\ \hline & b_1 & b_2 \end{array}$$

$$c_\ell = \sum_{j=1}^{s} a_{\ell j}$$

Method is $k_1 = f(t_i, y_i)$

$$k_2 = f(t_i + ah,\ y_i + ahk_1)$$

$$y_{i+1} = y_i + h(b_1 k_1 + b_2 k_2)$$

Plug in exact solution: $k_1 = f(t, y(t))$

$$k_2 = f(t + ah,\ y(t) + hak_1)$$

$$y(t+h) = \underbrace{y(t) + hb_1k_1 + hb_2k_2}_{(*)} + h\tau$$

Now we use Taylor's theorem:

Abbreviate $k_1 = f = f(t, y(t))$, $y(t) = y$

$$k_2 = f(t + ha,\ y(t) + hak_1)$$

$$= f + ha\frac{\partial f}{\partial t} + hak_1\frac{\partial f}{\partial y}(t, y(t)) + O(h^2)$$

$$= f + ha\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right) + O(h^2) \qquad (**)$$

On the other hand:

$$y(t+h) = y + hy' + \frac{h^2}{2}y'' + O(h^3)$$

$$= y + hf + \frac{h^2}{2}\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right) + O(h^3) \qquad (***)$$

we computed this last time

$$\Rightarrow \quad h\tau = y(t+h) - (*)$$

$$\underset{(***)}{=} y + hf + \frac{h^2}{2}\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right) + O(h^3) - (*)$$

$$\underset{(**)}{=} y + hf + \frac{h^2}{2}\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right) + O(h^3) - \left(y + hb_1 f + hb_2\left(f + ha\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right) + O(h^2)\right)\right)$$

$$= y - y + hf(1 - b_1 - b_2) + h^2\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right)\left(\frac{1}{2} - ab_2\right) + O(h^3)$$

$$\tau = (1 - b_1 - b_2)f + h\left(\frac{1}{2} - ab_2\right)\left(\frac{\partial f}{\partial t} + f\frac{\partial f}{\partial y}\right) + O(h^2)$$

So method is 1st order accurate if $b_1 + b_2 = 1$

& 2nd order accurate if also $ab_2 = \frac{1}{2}$

Previous examples: Midpoint: $a = \frac{1}{2}$ $b_1 = 0$, $b_2 = 1$

$0 + 1 = 1$ ✓

$\frac{1}{2} \cdot 1 = \frac{1}{2}$ ✓ $\Rightarrow$ 2nd order accurate

Modified Euler: $a = 1$, $b_1 = \frac{1}{2}$, $b_2 = \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2} = 1$ ✓

$1 \cdot \frac{1}{2} = \frac{1}{2}$ ✓ $\Rightarrow$ 2nd order accurate

Other important methods: Implicit method → Butcher tableau has nonzero entries on diagonal (&above)

→ they can be more stable than explicit methods

Important examples: Backward/implicit Euler: (1st order accurate)

$$y_{i+1} = y_i + hf(t_i + h, y_{i+1})$$

→ need Newton/fixed point/etc. method to find y_{i+1}

(implicit) midpoint rule (2nd order accurate

$$y_{i+1} = y_i + h f\left(t_i + \frac{h}{2}, \frac{y_i + y_{i+1}}{2}\right)$$

Trapezoidal rule (2nd order accurate)

$$y_{i+1} = y_i + \frac{h}{2}\left(f(t_i, y_i) + f(t_i + h, y_{i+1})\right)$$